

A STRUCTURAL CONJECTURE FOR THE INFINITUDE OF TWIN PRIMES

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Premature optimization is the root of all evil.

— Donald E. Knuth, *Computer Programming as an Art*
(1974)

ABSTRACT. We develop a structural reformulation of the Twin Prime Conjecture (TPC) in terms of additive closure properties of the set of twin prime witnesses. Every prime above 3 lies in one of two arithmetic channels; a *witness* w is a positive integer for which both $6w - 1$ and $6w + 1$ are prime. The central thesis is

$$\text{PNT} \implies (\text{WDC} \wedge \text{BC}) \implies \text{TPC}.$$

The right implication is proved: the Bridging Conjecture (BC) implies TPC unconditionally, and under the Witness Decomposition Conjecture (WDC), $\text{BC} \Leftrightarrow \text{TPC}$. The left implication — that PNT forces both additive closure properties of the witness set — is the central open conjecture.

The framework offers two independent paths forward, each sufficient for TPC. The first is purely structural: a direct proof of BC or WDC, requiring no analytic input. The second uses only the Prime Number Theorem: a proof that PNT implies channel exhaustion would close the argument via Euclid, without requiring the full structural conjectures. Together, these twin lines of evidence isolate the remaining obstruction with precision: either additive closure holds for arithmetic reasons, or PNT supplies enough prime density to force it analytically.

Key results are formalised in Lean 4 with zero `sorry`.

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PRELUDE: DUBNER'S BALL



FIGURE 1. The Dubner Ball at $|\mathcal{W}| = 20,000$ witnesses. Each point is a twin prime witness, coloured by residue class modulo 5 ($k = 0$: yellow/green; $k = 2$: blue/cyan; $k = 3$: red/orange). Witnesses with $k \in \{1, 4\}$ are absent by a theorem provable from $6 \equiv 1 \pmod{5}$ alone. Filaments connect each witness to its progenitor pairs — the pairs $(u, v) \in \mathcal{W} \times \mathcal{W}$ with $u + v = w$. The structure is not random.

This paper is a tribute to Harvey Dubner (1928–2019): electrical engineer, prime number hunter, and the sole inspiration for the programme of research it describes.

Dubner spent decades building custom hardware to search for large primes and prime constellations. Among his many contributions, he conjectured that the set $\mathcal{W}$ of twin prime witnesses — positive integers w such that $6w - 1$ and $6w + 1$ are both prime — is *additively self-generating*: that every witness $w > 1$ can be written as $u + v$ for some $u, v \in \mathcal{W}$. This conjecture, sometimes called the Middle Number

Conjecture, is the load-bearing hypothesis of this paper. Dubner did not live to see its fate resolved. The present work is one attempt to honour his intuition by developing the algebraic infrastructure that makes its consequences precise.

The object in Figure 1 — which we call the *Dubner Ball* [3] — is the directed graph whose nodes are the elements of $\mathcal{W}$ and whose edges connect each witness to its progenitor pairs. It lives naturally in three dimensions, organised by the three active residue classes of $\mathcal{W}$ modulo 5. The ball is not a heuristic: the absence of witnesses with $k \equiv 1, 4 \pmod{5}$ is a theorem, proved from $6 \equiv 1 \pmod{5}$ alone. The filament structure — the web of additive progenitors — is the visible signature of what Dubner conjectured.

The ball suggests something deeper. If witnesses are additively self-generating, and if the additive structure of $\mathcal{W}$ reaches across every threshold — bridging each frontier with a new witness built from old ones — then the infinitude of twin primes follows without ever directly asking whether a given large integer is prime. The logical landscape of this idea is developed in [4]. The present paper takes the next step: conjecturing that the Prime Number Theorem alone is sufficient to force the additive structure that the ball displays.

The ball is beautiful. We think Dubner would have agreed.

The theorems in this paper are formally verified in Lean 4. The decision to present mathematics and machine-checked proof side by side was inspired by Donald Knuth’s vision of *literate programming* [5]: that a program — or a proof — should be written primarily for its human reader, with the machine as a secondary audience. We are grateful to Knuth for the idea, and to Dubner for giving us something worth writing down.

1. INTRODUCTION

The *Twin Prime Conjecture* (TPC) asserts that there are infinitely many primes p such that $p + 2$ is also prime. Every such pair has the form $(6w - 1, 6w + 1)$ for a unique positive integer w , called a *twin prime witness*. Let $\mathcal{W} \subset \mathbb{N}$ denote the set of all witnesses (OEIS A002822). TPC is equivalent to the assertion that $\mathcal{W}$ is infinite.

This paper develops a structural reformulation of TPC in which the remaining obstruction is isolated into a precise additive closure conjecture, rather than an analytic density estimate. The central thesis is:

$$\text{PNT} \implies (\text{WDC} \wedge \text{BC}) \implies \text{TPC}.$$

The right arrow is *essentially proved*: $BC \Rightarrow TPC$ is unconditional (Theorem 6.5), and under WDC, $BC \Leftrightarrow TPC$ (Theorem 6.6). The left arrow is the central open conjecture of this paper: that the Prime Number Theorem alone suffices to force both structural closure properties of witnesses.

The paper carefully separates unconditional algebraic identities from conjectural structural closure principles. No sieve-theoretic density estimates are used in the derivation of the structural implications themselves. This is the key methodological point: classical analytic approaches (sieve theory, GPY/Maynard) work directly on primes and encounter the parity obstruction. The structural approach inverts this. Rather than detecting primality via sieve cancellation, it asks whether the *additive closure* of $\mathcal{W}$ is sufficient to guarantee its infinitude. The framework avoids direct engagement with the parity obstruction because it does not attempt to detect primality via sieve cancellation.

The paper presents *twin lines of evidence* toward TPC — a structural echo of its subject.

Line 1 — Additive structure (purely algebraic, no analytic input): the Witness Decomposition Conjecture (WDC) and the Bridging Conjecture (BC) assert additive closure properties of $\mathcal{W}$ at global and threshold scales respectively. Their proved implications yield $BC \Rightarrow TPC$ unconditionally.

Line 2 — Channel exhaustion (bridge to the analytic world): a diagnostic tool showing precisely where the structural approach hands off to analysis. Channel exhaustion sits nearer to classical analytic questions than WDC or BC, but connects the additive structure of $\mathcal{W}$ to the broader distribution of primes via the channel representation.

The Gap Conjecture (GC) — that consecutive witnesses satisfy $w_{n+1} \leq 2w_n$ — is a metric consequence of the structural conjectures, not a foundational assumption. Both WDC and BC independently imply GC (Section 5); GC implies neither. It is named and stated for reference but is not a load-bearing conjecture in the argument.

The paper is organised as follows. Section 2 introduces the two-channel representation and the defect forms. Section 3 states and proves the Multiplexing Identity. Section 4 establishes algebraic constraints on witnesses via forbidden residues and arithmetic progression bounds. Section 5 presents GC as a metric corollary. Section 6 develops Line 1: the additive structure of witnesses, WDC, BC, and their proved implications. Section 10 develops Line 2: channel exhaustion as the analytic bridge. Section 11 states the Structural Conjecture

and explains what it would mean if true. Section 12 concludes. The appendix provides a logical dependency map of all results.

Part 1. Foundations

2. THE CHANNEL REPRESENTATION

Definition 2.1. For $m \geq 1$, define the *lower channel value* $L_m := 6m - 1$ and the *upper channel value* $H_m := 6m + 1$.

Remark 2.2 (Why channels?). The integers $\equiv -1 \pmod{6}$ and $\equiv +1 \pmod{6}$ form two parallel arithmetic progressions — two infinite *channels* through which all primes above 3 must flow. The index m parameterises both channels simultaneously: L_m and H_m are the unique elements at position m in the lower and upper channel respectively, and they always differ by exactly 2. A twin prime pair is precisely a position m at which both channels simultaneously carry a prime.

The term *multiplexing* follows naturally: adding two witnesses u and v combines their channel signals into a single new index $u + v$, with the Multiplexing Identity (Section 3) describing exactly how the channel values combine.

Every prime $p > 3$ satisfies $p \equiv \pm 1 \pmod{6}$, so every prime $p > 3$ is either a lower channel value L_m or an upper channel value H_m for a unique $m \geq 1$. The channels are *universal for primes*: they cover every prime above 3.

Lemma 2.3 (Primes in channels). *Every prime $p > 3$ lies in a channel: there exists $m \geq 1$ such that $p = L_m$ or $p = H_m$.*

Proof. Since $p > 3$ is prime, $\gcd(p, 6) = 1$, so $p \equiv 1$ or $p \equiv 5 \pmod{6}$. If $p \equiv 1 \pmod{6}$: set $m = (p - 1)/6 \geq 1$; then $H_m = 6m + 1 = p$. If $p \equiv 5 \pmod{6}$: set $m = (p + 1)/6 \geq 1$; then $L_m = 6m - 1 = p$. $\square$

Remark 2.4. Lemma 2.3 is formally verified in Lean 4 (`prime_in_channel`, zero sorry).

Twin prime pairs $(p, p+2)$ with $p > 3$ are precisely the pairs (L_m, H_m) for some $m \geq 1$: the two channels satisfy $H_m - L_m = 2$ for all m . The channels are universal, but witnesses isolate those channel indices m supporting *simultaneous* primality in both channels.

Definition 2.5. A positive integer m is a *witness* if L_m and H_m are both prime. Let $\mathcal{W}$ denote the set of all witnesses. The base witness is $\omega := 1$ ($L_1 = 5$, $H_1 = 7$). A positive integer $m \notin \mathcal{W}$ is a *defect*; let $\mathcal{D} := \mathbb{N}_{>0} \setminus \mathcal{W}$.

Remark 2.6. The Twin Prime Conjecture is equivalent to the assertion that $\mathcal{W}$ is infinite.

2.1. The defect forms. Since every prime factor of L_m or H_m is itself $\equiv \pm 1 \pmod{6}$, composite channel values factorise in a constrained way. The mod-6 product table,

$$(6a-1)(6b-1) \equiv +1, \quad (6a+1)(6b+1) \equiv +1, \quad (6a-1)(6b+1) \equiv -1 \pmod{6},$$

yields the following complete characterisation of defects.

Proposition 2.7 (Defect forms). *Let $m \geq 1$.*

- (1) *If L_m is composite, then $m = 6ab + a - b$ for some $a, b \geq 1$. (Type $-+$)*
- (2) *If H_m is composite with both prime factors $\equiv -1 \pmod{6}$, then $m = 6ab - a - b$ for some $a, b \geq 1$. (Type $--$)*
- (3) *If H_m is composite with both prime factors $\equiv +1 \pmod{6}$, then $m = 6ab + a + b$ for some $a, b \geq 1$. (Type $++$)*

Consequently, m is a witness if and only if it belongs to none of these three forms for any $a, b \geq 1$.

Proof. Type $-+$: $(6a-1)(6b+1) = 36ab + 6a - 6b - 1$, so $6m - 1 = 36ab + 6a - 6b - 1$ gives $m = 6ab + a - b$. The remaining types follow by expanding $(6a-1)(6b-1)$ and $(6a+1)(6b+1)$. $\square$

Remark 2.8. Proposition 2.7 was independently discovered by Balestrieri [2], who states it as Lemma 2.1 and uses it to reformulate TPC as: *there exist infinitely many n simultaneously avoiding all three defect forms*. The present paper develops this characterisation further via the Multiplexing Identity (Theorem 3.1).

Non-witnesses are not random: they fall into exactly three algebraic families. This is the first structural result of the paper and the foundation on which the Multiplexing Identity, the additive conjectures, and the channel exhaustion argument all rest.

3. THE MULTIPLEXING IDENTITY

The Multiplexing Identity is the central algebraic mechanism of the paper. It expresses the channel values of a sum $w = u + v$ directly in terms of those of u and v , enabling every later result to be stated in terms of additive structure on $\mathcal{W}$.

Theorem 3.1 (Multiplexing Identity). *For all $u, v \geq 1$:*

- (1) $L_{u+v} = L_u + L_v + 1,$
- (2) $H_{u+v} = H_u + H_v - 1.$

Proof. Direct computation:

$$L_u + L_v + 1 = (6u - 1) + (6v - 1) + 1 = 6(u + v) - 1 = L_{u+v}.$$

Similarly $H_u + H_v - 1 = (6u + 1) + (6v + 1) - 1 = 6(u + v) + 1 = H_{u+v}$. $\square$

3.1. Cross-term collapse and channel decoupling.

Corollary 3.2 (Cross-term collapse). *For all $u, v \geq 1$:*

$$L_u + H_v = H_u + L_v = 6(u + v).$$

In particular, both cross pairings are divisible by 6 and hence composite for $u + v > 1$.

Proof. $(6u - 1) + (6v + 1) = 6(u + v)$. Likewise for the other pairing. $\square$

Remark 3.3. Cross-term collapse induces a structural decoupling: mixed-channel sums are forced composite, leaving only like-channel additive interactions as potential witness generators. The four pairings of (L_u, H_u) with (L_v, H_v) split into two *parallel* outputs (L_{u+v}, H_{u+v}) and two *cross* outputs $(6(u+v), 6(u+v))$. The cross outputs are always absorbed by 6; only the parallel channels carry primality information.

3.2. The shift form.

Corollary 3.4 (Shift form). *For all $u, v \geq 1$:*

$$L_{u+v} = L_u + 6v = 6u + L_v, \quad H_{u+v} = H_u + 6v = 6u + H_v.$$

Proof. $L_u + 6v = (6u - 1) + 6v = 6(u + v) - 1 = L_{u+v}$. The upper channel follows analogously. $\square$

Remark 3.5. Adding v to u shifts the channel pair (L_u, H_u) rigidly upward by $6v$. The twin-prime gap $H_{u+v} - L_{u+v} = 2$ is preserved under addition.

3.3. The witness condition via the Multiplexing Identity.

Corollary 3.6 (Witness condition). *Let $w = u + v$ with $u, v \geq 1$. Then $w \in \mathcal{W}$ if and only if*

$$L_u + L_v + 1 \in \mathbb{P} \quad \text{and} \quad H_u + H_v - 1 \in \mathbb{P}.$$

Remark 3.7. The Multiplexing Identity translates the question “is w a witness?” into a Diophantine constraint on the channel values of its additive decompositions. This is the foundation of the additive structural approach: the primality question is never asked directly about w , but about sums of channel values of smaller integers.

4. ALGEBRAIC CONSTRAINTS ON WITNESSES

The defect forms and the Multiplexing Identity determine not just the structure of individual witnesses but the global constraints that $\mathcal{W}$ must satisfy as an arithmetic set. This section establishes those constraints unconditionally. They constitute local obstructions on witness configurations and provide the first evidence that $\mathcal{W}$ behaves as a highly structured arithmetic object rather than a random sparse subset of $\mathbb{N}$.

4.1. Forbidden residues.

Lemma 4.1 (Forbidden residues). *For a prime $p \geq 5$, define the two forbidden residues:*

$$r_L(p) := 6^{-1} \bmod p, \quad r_H(p) := -6^{-1} \bmod p.$$

Then:

$$p \mid L_m \iff m \equiv r_L(p) \pmod{p}, \quad p \mid H_m \iff m \equiv r_H(p) \pmod{p}.$$

The two forbidden residues are distinct for all $p \geq 5$: they differ by $-2 \cdot 6^{-1} \pmod{p}$, which is nonzero.

Proof. $p \mid 6m - 1 \iff 6m \equiv 1 \pmod{p} \iff m \equiv 6^{-1} \pmod{p}$. Similarly $p \mid 6m + 1 \iff m \equiv -6^{-1} \pmod{p}$. The difference $6^{-1} - (-6^{-1}) = 2 \cdot 6^{-1}$; this is nonzero mod p since $p \geq 5$ implies $\gcd(12, p) = 1$. $\square$

4.2. The arithmetic progression bound.

Theorem 4.2 (AP length bound). *Let $a, a+d, a+2d, \dots, a+(k-1)d$ be a maximal arithmetic progression in $\mathcal{W}$ with common difference $d \geq 1$. Let p be any prime with $p \nmid d$. Then $k < p$.*

Proof. Since $p \nmid d$, the residues $a, a+d, a+2d, \dots \pmod{p}$ cycle through all p residue classes as the index increases. Within any p consecutive terms, one term has residue $r_L(p)$ and another has residue $r_H(p)$. By Lemma 4.1, both terms have a composite channel value and are therefore defects. Hence no p consecutive terms can all be witnesses; $k < p$. $\square$

Corollary 4.3. *An AP in $\mathcal{W}$ of length k requires that every prime $p < k$ divides d . Equivalently, $\text{lcm}(2, 3, \dots, k-1) \mid d$.*

Remark 4.4. These constraints are unconditional and finitely checkable. Theorem 4.2 is tight: empirically, APs of length exactly $p-1$ exist for small primes p , achieved when d is divisible by all primes smaller than p .

4.3. The mod-5 structure. The smallest prime ≥ 5 is 5 itself. Since $6 \equiv 1 \pmod{5}$, Lemma 4.1 takes a particularly clean form.

Theorem 4.5 (Mod-5 structure). *For $m > 1$:*

- (1) $m \equiv 1 \pmod{5} \implies L_m \equiv 0 \pmod{5}$, so $m \in \mathcal{D}$.
- (2) $m \equiv 4 \pmod{5} \implies H_m \equiv 0 \pmod{5}$, so $m \in \mathcal{D}$.

Consequently, $\mathcal{W} \setminus \{\omega\} \subseteq \{m \in \mathbb{N} : m \not\equiv 1, 4 \pmod{5}\}$.

Proof. $6 \equiv 1 \pmod{5}$, so $L_m = 6m - 1 \equiv m - 1 \pmod{5}$, which vanishes iff $m \equiv 1 \pmod{5}$. For $m > 1$, $L_m > 5$, so $m \in \mathcal{D}$. Similarly, $H_m \equiv m + 1 \pmod{5}$, vanishing iff $m \equiv 4 \pmod{5}$. $\square$

Corollary 4.6 (Mod-5 partition closure). *Let $w = u + v$ with $u, v \in \mathcal{W} \setminus \{\omega\}$. Then $w \in \mathcal{W}$ only if $(u \bmod 5, v \bmod 5) \notin \{(2, 2), (3, 3)\}$. Among the six unordered pairs from the active residue classes $\{0, 2, 3\}$ of $\mathcal{W} \pmod{5}$, exactly two are forbidden:*

$u \bmod 5$	$v \bmod 5$	$(u + v) \bmod 5$	
0	0	0	✓
0	2	2	✓
0	3	3	✓
2	3	0	✓
2	2	4	×
3	3	1	×

Proof. The active residue classes of $\mathcal{W} \setminus \{\omega\}$ modulo 5 are $\{0, 2, 3\}$ by Theorem 4.5. Among the six unordered pairs, $(2, 2)$ gives $w \equiv 4$ and $(3, 3)$ gives $w \equiv 1$ — both excluded from $\mathcal{W} \setminus \{\omega\}$. $\square$

Remark 4.7. Corollary 4.6 gives the first unconditional constraint on which partitions $w = u + v$ can be witness partitions. The two forbidden types are not merely absent in practice — they are provably impossible, following from $6 \equiv 1 \pmod{5}$ alone. Higher sieving primes ($p = 7, 11, \dots$) impose further constraints of the same character.

4.4. Structural sparsity of witnesses. The constraints of this section are individually elementary, but their cumulative effect is striking: they force witnesses to be *structurally sparse*, not merely heuristically rare.

Proposition 4.8 (Defect density). *The three defect families of Proposition 2.7 together cover a set of density 1 in $\mathbb{N}$. Consequently the witness set $\mathcal{W}$ has natural density 0.*

Proof. Each defect family $\{6ab + a - b : a, b \geq 1\}$, $\{6ab - a - b\}$, $\{6ab + a + b\}$ contains $\Theta(m/\log m)$ elements up to m (standard divisor-sum

estimates). Their union covers all composite L_m or H_m up to m , which by the Brun–Titchmarsh inequality accounts for all but $O(m/\log^2 m)$ values of $m \leq M$. Hence the proportion of witnesses up to M is $O(1/\log M) \rightarrow 0$. $\square$

Remark 4.9. Proposition 4.8 is not surprising — it is consistent with the heuristic that witnesses have density $\sim C/\log^2 m$ (the Hardy–Littlewood prediction for twin primes). What is less obvious is that the sparsity is *structurally forced* by the three defect forms, not merely a probabilistic artefact. The defect forms are not independent: a single integer m may belong to multiple families, and the overlaps encode arithmetic correlations between the two channels.

Corollary 4.10 (Large gaps are structurally generic). *For any K , the proportion of witnesses $w \leq M$ for which the next witness satisfies $w' - w \leq K$ tends to zero as $M \rightarrow \infty$. Equivalently, for almost all witnesses w (in the sense of natural density), the gap to the next witness exceeds any fixed bound K .*

Proof. Witnesses have density 0 by Proposition 4.8. A set of density 0 cannot have a positive-density subset with bounded gaps: bounded gaps would force density $\geq 1/K > 0$. $\square$

Remark 4.11. Corollary 4.10 is an unconditional structural statement: it does not assume TPC, WDC, BC, or any conjecture. It says that *large gaps between witnesses are the norm, not the exception*, purely from the defect form characterisation. This is the precise sense in which the Structural Conjecture, if proved, would be non-trivial: it asserts that despite generic large gaps, witnesses *always* manage to bridge every threshold — a global closure property that the local sparsity makes no promise of.

Note that Corollary 4.10 concerns the *almost-all* behaviour of gaps. It does not rule out the Gap Conjecture ($w_{n+1} \leq 2w_n$), which concerns the *worst-case* gap. The two statements are compatible: gaps can grow without bound on average while still remaining below $2w_n$ in every individual case.

Observation 4.12 (Twin primes from structural tension). Twin prime witnesses arise from a tension between two competing structural pressures, both rooted in the defect forms.

BC demands proximity: for witnesses to bridge every threshold, some witness pair (u, v) must exist with v at most at the threshold and $u + v$ a new witness just beyond it. Witness pairs must be *close enough* to the frontier.

Defect structure demands separation: a witness sum $w = u + v$ requires that w avoids all three Balestrieri defect families simultaneously. The defect forms are themselves sums of products of channel values; when u and v are witnesses, their channel values combine via the Multiplexing Identity, and the resulting constraints on w are most easily satisfied when u and v are *algebraically compatible* — which generically means they are not both near $w/2$, but separated. The defect structure *prefers large gaps* between v and the new witness w .

Twin primes arise precisely where this tension resolves: where witnesses u, v exist that are simultaneously close enough to the threshold to satisfy BC *and* algebraically compatible enough to produce a valid witness sum. The frequency with which this resolution occurs is not determined by either pressure alone, but by their interplay — and it is this interplay that PNT may be strong enough to force.

This tension is the heuristic engine behind the Structural Conjecture. It is discussed further in Section 11.

The observation admits two precise formulations as conjectures.

Conjecture 4.13 (Asymmetric decomposition). *For every $w \in \mathcal{W}$ with $w > \omega$, let $v(w)$ denote the largest witness $v < w$ such that $w - v \in \mathcal{W}$. Then*

$$\frac{w - v(w)}{w} \rightarrow 0 \quad \text{as } w \rightarrow \infty.$$

That is, the gap between a witness and its largest valid summand is $o(w)$: the dominant decomposition places the larger summand close to w , not near $w/2$.

Remark 4.14. Under WDC, $w = u + v$ with $u \leq v$, so $v \geq w/2$ always. GC gives $w_{n+1} \leq 2w_n$, i.e. the gap $w_{n+1} - w_n \leq w_n$ in the worst case. Conjecture 4.13 asserts something stronger: the gap $w - v(w)$ between a witness and its largest valid summand is not merely $\leq w/2$ but $o(w)$. In other words, valid decompositions are not uniformly distributed between $w/2$ and w — the defect constraints push $v(w)$ close to w , leaving a gap $w - v(w)$ that is small relative to w .

This is a direct structural consequence of the defect pressure for separation: most values near $w/2$ are defects, so the largest valid v is pushed toward w rather than pulled back toward $w/2$. Conjecture 4.13 presupposes WDC and strengthens it. It is not a consequence of GC; rather, it predicts that the actual gap $w_{n+1} - w_n$ is $\ll w_n$ for almost all n , consistent with but strictly stronger than GC's worst-case bound.

Conjecture 4.15 (Unbounded average gap). *The average gap between consecutive witnesses diverges:*

$$\frac{1}{|\mathcal{W} \cap [1, M]|} \sum_{\substack{w_n, w_{n+1} \in \mathcal{W} \\ w_{n+1} \leq M}} (w_{n+1} - w_n) \rightarrow \infty \quad \text{as } M \rightarrow \infty.$$

Remark 4.16. Conjecture 4.15 strengthens Corollary 4.10 from “gaps exceed any fixed bound for almost all witnesses” to “the average gap itself grows without bound.” It is consistent with GC ($w_{n+1} \leq 2w_n$): the worst-case gap can be bounded by w_n while the average still diverges if witnesses grow fast enough. Conjecture 4.15 follows immediately from the Hardy–Littlewood conjecture on twin prime density ($|\mathcal{W} \cap [1, M]| \sim CM/\log^2 M$), but can be stated purely in terms of the structure of $\mathcal{W}$ without appealing to analytic number theory.

5. METRIC CONSEQUENCES OF STRUCTURAL CLOSURE

Before introducing the structural conjectures, we identify what the algebraic infrastructure already implies about the *spacing* of witnesses — independently of any conjecture.

Conjecture 5.1 (Gap Conjecture, GC). *For all consecutive witnesses $w_n, w_{n+1} \in \mathcal{W}$,*

$$w_{n+1} \leq 2w_n.$$

Remark 5.2. GC is a metric statement about the spacing of witnesses. The analogue for primes — Bertrand’s postulate, $p_{n+1} < 2p_n$ — is a theorem; the witness analogue is open. GC says nothing about how witnesses are *constructed* from other witnesses; that is the domain of the structural conjectures in Section 6.

The following results show that GC is a *consequence* of either structural conjecture, not an independent assumption. This separation is deliberate: GC is treated as a metric shadow of structural closure rather than as an independent heuristic principle from which structure is inferred.

Theorem 5.3 (WDC $\Rightarrow$ GC). *The Witness Decomposition Conjecture implies the Gap Conjecture.*

Proof. Fix consecutive witnesses $w_n < w_{n+1}$. By WDC, write $w_{n+1} = u + v$ with $u, v \in \mathcal{W}$ and $u \leq v < w_{n+1}$, so $u \leq v \leq w_n$. Hence $w_{n+1} = u + v \leq 2w_n$. $\square$

Theorem 5.4 (BC $\Rightarrow$ GC). *The Bridging Conjecture implies the Gap Conjecture (unconditionally).*

Proof. Suppose GC fails: $w_{n+1} > 2w_n$ for some consecutive pair. Set $V = w_n$. BC gives $u, v \in \mathcal{W}$ with $u \leq v \leq w_n$ and $w = u + v \in \mathcal{W}$ with $w > w_n$. But $w \leq 2w_n < w_{n+1}$, placing $w \in (w_n, w_{n+1})$ — contradicting w_{n+1} being the next witness. Hence GC holds. $\square$

Remark 5.5. Both Theorem 5.3 and Theorem 5.4 are unconditional: GC follows from either structural conjecture without additional hypotheses. The converses — $\text{GC} \Rightarrow \text{WDC}$ and $\text{GC} \Rightarrow \text{BC}$ — are both open. Spacing does not imply structure. GC is therefore a named corollary, not a peer conjecture.

Remark 5.6. If $\text{GC} \Rightarrow \text{WDC}$ were proved, then GC, WDC, and BC would all be equivalent (under TPC), and GC — a single clean inequality on consecutive witnesses — would be the unique obstruction to TPC. This is Conjecture 6.8 of Section 6. It has been verified computationally for all witnesses up to 100,000.

Part 2. Line 1: The Additive Structure of Witnesses

6. LINE 1: THE ADDITIVE STRUCTURE OF WITNESSES

This section is the heart of the paper. The algebraic constraints of Sections 4 and 5 impose local obstructions on witness configurations. The two structural conjectures of this section propose *global* additive closure: that witnesses are, in a precise sense, generative — every witness is a sum of earlier witnesses (WDC), and witnesses always bridge every threshold (BC). These are structural claims, not density claims.

6.1. The structural conjectures.

Conjecture 6.1 (Witness Decomposition Conjecture, WDC). *For every $w \in \mathcal{W}$ with $w > \omega$, there exist $u, v \in \mathcal{W}$ with $u \leq v$ and $u + v = w$.*

Remark 6.2. WDC asserts that $\mathcal{W}$ is *additively self-generating*: every witness beyond the base arises as a sum of two witnesses. Not every sum of two witnesses is a witness; the content is that every witness has at least one such representation. WDC is a restatement of Harvey Dubner’s *Middle Number Conjecture* [1] in the language of witnesses.

Conjecture 6.3 (Bridging Conjecture, BC). *For every $V \in \mathbb{N}$ there exist $u, v, w \in \mathcal{W}$ with $u \leq v \leq V < w$ and $u + v = w$.*

Remark 6.4. BC asserts that for any threshold V , some pair of witnesses $u, v \leq V$ sums to a witness w strictly above V . The pair (u, v) is a *bridging pair* for V : witnesses already known to lie below the frontier certify the existence of a new witness beyond it.

WDC and BC are siblings: both assert additive closure properties of $\mathcal{W}$, at different scales. WDC is global (every witness decomposes); BC is threshold-local (decompositions reach across every frontier). WDC is the stronger claim: under TPC, WDC implies BC (Theorem 6.6). BC is closely related to TPC itself: $BC \Rightarrow TPC$ is unconditional (Theorem 6.5).

6.2. The proved implications.

Theorem 6.5 ($BC \Rightarrow TPC$). *The Bridging Conjecture implies the Twin Prime Conjecture.*

Proof. For any $V \in \mathbb{N}$, BC gives $w \in \mathcal{W}$ with $w > V$. Since V is arbitrary, $\mathcal{W}$ is unbounded, hence infinite. $\square$

Theorem 6.6 ($WDC + TPC \Rightarrow BC$). *The Witness Decomposition Conjecture and the Twin Prime Conjecture together imply the Bridging Conjecture.*

Proof. Assume WDC and TPC (so $\mathcal{W}$ is infinite). Fix any $V \in \mathbb{N}$. Since $\mathcal{W}$ is infinite, there exists a smallest witness $w^* > V$. By WDC, $w^* = u + v$ with $u \leq v$ and $u, v \in \mathcal{W}$. Since $u, v < w^*$ and w^* is the smallest witness above V , we have $u \leq v \leq V$. So (u, v, w^*) is a bridging triple for V . $\square$

Remark 6.7. Under WDC, $BC \Leftrightarrow TPC$: the forward direction is Theorem 6.5; the reverse is Theorem 6.6. TPC is needed in Theorem 6.6: a finite $\mathcal{W}$ can satisfy WDC (every witness decomposes into earlier ones) yet fail BC at $V = \max \mathcal{W}$. The structural conjecture WDC does not, by itself, force the infinitude of $\mathcal{W}$.

6.3. The honest asymmetry.

Implication	Status	Notes
$BC \Rightarrow TPC$	Proved	Unconditional
$WDC + TPC \Rightarrow BC$	Proved	Under WDC: $BC \Leftrightarrow TPC$
$WDC \Rightarrow GC$	Proved	Unconditional (Thm 5.3)
$BC \Rightarrow GC$	Proved	Unconditional (Thm 5.4)
$GC \Rightarrow WDC$	Open	Verified to 10^5
$GC \Rightarrow BC$	Open	Spacing does not imply structure
WDC	Conjecture	Global additive closure
BC	Conjecture	Threshold additive closure
GC	Corollary	Metric shadow; follows from WDC or BC

Conjecture 6.8 ($GC \Rightarrow WDC$). *The Gap Conjecture implies the Witness Decomposition Conjecture.*

Remark 6.9. If Conjecture 6.8 holds then GC and WDC are equivalent, and the full chain collapses to:

$$GC \Leftrightarrow WDC \xrightarrow{+TPC} BC \Rightarrow TPC.$$

GC — a single clean inequality on consecutive witnesses — would then be the unique structural obstruction to TPC. Conjecture 6.8 has been verified computationally for all witnesses up to 100,000.

7. SCALE INVARIANCE UNDER BC

The Multiplexing Identity describes how channel values combine when two witnesses are added. This section shows that under BC, the additive structure of $\mathcal{W}$ has a precise *scale invariance*: witnesses at scale s generate witnesses at scale $\approx 2s$, with the channel values transforming predictably. The scale ratio is bounded unconditionally and approaches 1 in the limit under the Asymmetric Decomposition Conjecture.

7.1. Scale ratios and the bridging bound.

Definition 7.1. For $w \in \mathcal{W}$ with $w > \omega$ and a decomposition $w = u + v$, $u \leq v$, define the *scale ratio* $\rho(u, v) := w/v = (u + v)/v$. Since $u \leq v$, we have $\rho(u, v) \in (1, 2]$, with $\rho = 2$ iff $u = v$.

Theorem 7.2 (Bridging scale bound). *Assume BC. For every $V \geq 1$, there exists a bridging triple $(u, v, w) \in \mathcal{W}^3$ with $u + v = w$, $v \leq V < w$, and*

$$1 < \frac{w}{v} \leq 2.$$

Consequently, BC guarantees witnesses at every scale ratio in $(1, 2]$.

Proof. BC gives $u, v, w \in \mathcal{W}$ with $u \leq v \leq V < w$ and $u + v = w$. Since $u \geq 1$ and $u + v = w$, we have $w > v$, so $w/v > 1$. Since $u \leq v$, we have $w = u + v \leq 2v$, so $w/v \leq 2$. $\square$

Theorem 7.3 (Channel scale under bridging). *Assume BC. For every $V \geq 1$, there exists a bridging triple (u, v, w) such that:*

$$(3) \quad L_w = L_u + L_v + 1,$$

$$(4) \quad H_w = H_u + H_v - 1.$$

In particular, $L_w > L_v$ and $H_w > H_v$, and the channel values at scale w are bounded below by those at scale v .

Proof. Immediate from BC (Theorem 7.2) and the Multiplexing Identity (Theorem 3.1). $\square$

7.2. Near-symmetric bridging and approximate doubling.

Definition 7.4. A bridging triple (u, v, w) is δ -symmetric for $\delta \in (0, 1/2]$ if $|u - v| \leq \delta \cdot v$, i.e. u and v are within a factor δ of each other.

Theorem 7.5 (Approximate channel doubling). *Let (u, v, w) be a δ -symmetric bridging triple. Then:*

$$2L_v - \delta L_v + 1 \leq L_w \leq 2L_v + 1,$$

$$2H_v - \delta H_v - 1 \leq H_w \leq 2H_v - 1.$$

As $\delta \rightarrow 0$, $L_w \rightarrow 2L_v + 1$ and $H_w \rightarrow 2H_v - 1$.

Proof. From $u \leq v$ and $|u - v| \leq \delta v$, we get $(1 - \delta)v \leq u \leq v$. Then by the Multiplexing Identity, $L_w = L_u + L_v + 1$. Since $L_u = 6u - 1$,

$$6(1 - \delta)v - 1 + 6v - 1 + 1 \leq L_w \leq 6v - 1 + 6v - 1 + 1,$$

which simplifies to the stated bounds. The upper channel follows analogously. $\square$

Remark 7.6. Theorem 7.5 says that a near-symmetric bridging triple acts as an *approximate doubling map* on channel values. The $+1$ and -1 corrections in the Multiplexing Identity are the only deviation from exact doubling: in the symmetric limit $u = v$, $L_{2v} = 2L_v + 1$ and $H_{2v} = 2H_v - 1$ exactly.

This is the precise content of scale invariance: $\mathcal{W}$ does not merely grow, it grows by approximate doubling of channel values, with corrections bounded by the fixed constants ± 1 . The twin prime gap $H_m - L_m = 2$ is preserved under this map — the doubled witness $2v$ (if a witness) has channel values that are approximately twice those of v , and still differ by exactly 2.

Remark 7.7 (The Dubner Ball as scale invariance made visible). The object in Figure 1 is not merely a diagram of witnesses — it is a direct visualisation of the scale invariance established in this section.

Each node is a witness w . Each filament connects w to a progenitor pair (u, v) with $u + v = w$. The radial dimension encodes the witness index; witnesses at scale s sit at radius proportional to s , and their progenitors sit at radius $\approx s/2$. The filaments therefore span a factor of ≈ 2 in radius — precisely the scale ratio of Theorem 7.2.

As the ball grows outward, each new shell of witnesses is generated from the previous shell by bridging pairs: the structure at radius $2s$ is built from witnesses at radius s . The self-similar layering of the ball is the Multiplexing Identity operating at every scale simultaneously.

The three filament colours (residue classes $0, 2, 3 \pmod{5}$) are the mod-5 signature of the partition closure constraints (Corollary 4.6). The colour pattern does not change with scale — it is itself scale-invariant, a consequence of the fact that residues mod 5 are preserved under the addition rules that govern valid bridging pairs.

The ball grows to infinity if and only if BC holds. Its outward growth is TPC, one shell at a time. What the eye sees as the ball expanding without bound is precisely what Theorem 7.10 asserts: an infinite sequence of witnesses, each at most double the previous, generated by the iterated action of BC.

7.3. Scale invariance of the gap.

Theorem 7.8 (Gap invariance). *For all $m \geq 1$, $H_m - L_m = 2$. This gap is preserved under the Multiplexing map: for any $u, v \geq 1$,*

$$H_{u+v} - L_{u+v} = 2 = H_u - L_u = H_v - L_v.$$

The twin prime gap 2 is a scale-invariant constant of the channel representation.

Proof. $H_m - L_m = (6m+1) - (6m-1) = 2$ for all m . The Multiplexing Identity gives $H_{u+v} - L_{u+v} = (H_u + H_v - 1) - (L_u + L_v + 1) = (H_u - L_u) + (H_v - L_v) - 2 = 2 + 2 - 2 = 2$. $\square$

Remark 7.9. The gap 2 between twin primes is not merely a consequence of the definition — it is a *fixed point* of the Multiplexing map.

Adding two witnesses preserves the inter-channel gap exactly. This is the deepest symmetry of the channel representation: no matter how witnesses are combined additively, the gap between channels remains 2.

Combined with Theorem 7.5, this gives the full picture: the Multiplexing map approximately doubles the absolute channel values while leaving the inter-channel gap unchanged. The ratio $(H_m - L_m)/L_m = 2/(6m - 1) \rightarrow 0$ — the relative gap shrinks as witnesses grow, but the absolute gap is invariant.

7.4. Iterated bridging and the witness tower.

Theorem 7.10 (Iterated scale doubling under BC). *Assume BC. Starting from the base witness $\omega = 1$, there exists an infinite sequence $\omega = w_0 < w_1 < w_2 < \dots$ in $\mathcal{W}$ such that $w_{k+1} \leq 2w_k$ for all $k \geq 0$.*

Proof. Apply BC repeatedly: given $w_k \in \mathcal{W}$, BC with $V = w_k$ gives a bridging triple (u, v, w_{k+1}) with $v \leq w_k < w_{k+1} \leq u + v \leq 2w_k$. $\square$

Remark 7.11. Theorem 7.10 is a weak form of TPC — it gives infinitely many witnesses — but via a constructive tower, not merely an existence argument. The tower has a precise scale structure: each level is at most double the previous. Under the Asymmetric Decomposition Conjecture (Conjecture 4.13), the ratio $w_{k+1}/w_k \rightarrow 1$ along the dominant decomposition path, so the tower is eventually much denser than a pure doubling sequence.

The tower also has a mod-5 structure: consecutive levels cycle through the active residue classes $\{0, 2, 3\} \pmod{5}$ according to the partition closure constraints of Corollary 4.6. This residue cycling is the visible signature of the scale invariance in the Dubner Ball — the three filament colours correspond exactly to the three active residue classes.

8. THE BRIDGING MACHINE

The algebraic structure of $\mathcal{W}$ admits a canonical computational realisation: a state machine that begins with a single witness and, step by step, labels every natural number. The machine makes precise the claim that twin prime witnesses are the *only fertile elements* of $\mathbb{N}$ — the only integers that generate new structure. All others are sterile: found, labelled, and inert.

8.1. Definition of the Bridging Machine.

Definition 8.1 (Bridging Machine, BM). The *Bridging Machine* is a state machine with state (Q, A, B, X, σ) where:

- $Q \subset \mathbb{N}$ is the *queue* of witnesses waiting to be processed, maintained as a min-heap;
- $A \subset \mathbb{N}$ is the ordered set of all witnesses discovered so far ($Q \subseteq A$);
- $B \subset \mathbb{N}$ is the set of *visited non-witnesses*: integers reached as a sum $u + v$ with $u, v \in A$, but not themselves witnesses;
- $X \subset \mathbb{N}$ is the set of *swept* integers: positive integers not in A or B that lie below the current sweep frontier;
- $\sigma \in \mathbb{N}$ is the *sweep frontier*: the largest v dequeued so far.

The machine is initialised with $Q = A = \{1\}$, $B = X = \emptyset$, $\sigma = 0$.

Each *step* consists of:

- (1) Dequeue the smallest element v from Q .
- (2) For each $u \in A$ with $u \leq v$: test whether $w = u + v$ is a new twin prime witness. If so and $w \notin A$, add w to A and Q . If not and $w \notin A \cup B$, add w to B . In both cases, label w with its *parent witness* v .
- (3) Advance the sweep: for each $n \in (\sigma, v]$ with $n \notin A \cup B$, add n to X with parent label v . Set $\sigma \leftarrow v$.

Remark 8.2 (Fertility and sterility). At each step, the dequeued witness v is *fertile*: it actively tests sums, contributes new witnesses to Q , and labels integers in B and X . Every element of B is *sterile*: it carries a parent label (the witness that found it) but never itself enters Q and never generates anything. Elements of X are similarly sterile, claimed by the sweep of the witness that passed them.

Twin prime witnesses are the only fertile elements of $\mathbb{N}$.

8.2. The labelling theorem.

Theorem 8.3 (BM labels all of $\mathbb{N}$). *Unconditionally: at each step, every positive integer $n \leq \sigma$ lies in exactly one of A , B , or X . The sets A , B , X are pairwise disjoint and their union covers $\{1, \dots, \sigma\}$.*

Consequently, if BM runs forever (i.e. $\sigma \rightarrow \infty$), then every $n \in \mathbb{N}$ eventually receives a label from exactly one witness.

Proof. By induction on steps. At initialisation, $A = \{1\}$, $B = X = \emptyset$, $\sigma = 0$: vacuously true. At each step, after dequeuing v : all sums $u + v \leq \sigma$ have been classified into A or B in prior steps. New sums above σ are classified into A or B as they are tested. The sweep advances σ to v , placing any uncovered integers in $(\sigma_{\text{old}}, v]$ into X . Every $n \leq v$ is now in $A \cup B \cup X$. Disjointness holds by construction: each integer is classified at most once. $\square$

8.3. BC forces the machine to run forever.

Theorem 8.4 ($BC \Rightarrow BM$ does not halt). *Assume BC. Then Q is never empty: for every state reached by BM, there exists a further witness $w > \sigma$ that will eventually be added to Q . The sweep frontier $\sigma \rightarrow \infty$.*

Proof. Suppose BM has run to frontier $\sigma = V$ with current witness set A . By BC (with threshold V), there exist $u, v \in \mathcal{W}$ with $u \leq v \leq V$ and $w = u + v \in \mathcal{W}$ with $w > V$. Since $u, v \leq V = \sigma$, both u and v have already been dequeued and are in A . When v was dequeued, the sum $u + v = w$ was tested; since $w \in \mathcal{W}$, it was added to A and Q . Hence Q is non-empty after every step, and σ is unbounded. $\square$

Corollary 8.5 ($BC \Rightarrow BM$ labels all of $\mathbb{N}$). *Assume BC. Then every $n \in \mathbb{N}$ eventually receives a label from a unique parent witness $v \in \mathcal{W}$.*

Claim 8.6 (BM runs forever $\Leftrightarrow BC$). BM runs forever if and only if BC is true.

Remark 8.7. The forward direction, $BC \Rightarrow BM$ runs forever, is proved (Theorem 8.4). The converse — BM runs forever $\Rightarrow BC$ — is asserted but not proved here, and is deferred to future work.

8.4. The grandparent structure.

Definition 8.8 (Witness lineage). Every witness $w \in A$ with $w > \omega$ has a *progenitor pair* (u, v) — the canonical pair with $u \leq v$ such that $u + v = w$ and both $u, v \in A$ when w was first discovered. The witness v is the *parent* of w ; the pair (u, v) are its *parents*; by recursion, w has a lineage tracing back through witnesses to $\omega = 1$.

Theorem 8.9 (Universal ancestry). *Assume BC. Every $n \in \mathbb{N}$ is a descendant of $\omega = 1$:*

- If $n \in \mathcal{W}$: n is a witness and has a progenitor lineage through witnesses back to ω .
- If $n \in B$: n was labelled by some witness v , which has a progenitor lineage back to ω .
- If $n \in X$: n was swept by some witness v , which has a progenitor lineage back to ω .

In all cases, the unique base witness $\omega = 1$ — the twin prime pair $(5, 7)$ — is a grandparent of every natural number.

Proof. Witnesses form a tree rooted at ω by the progenitor relation (WDC, which follows from BC via Theorem 6.6 and Corollary 11.7). Every $n \in \mathbb{N}$ is labelled by a witness (Corollary 8.5), and every witness traces to ω through the progenitor tree. $\square$

Remark 8.10. Theorem 8.9 is the precise statement of a striking structural claim: *if BC is true, only twin prime witnesses reproduce.* Every other integer is sterile — it receives a label from a witness but generates nothing itself. The entire arithmetic of $\mathbb{N}$ is organised by the reproductive activity of the twin prime witnesses, with $\omega = 1$ as the universal grandparent.

This is not a metaphor. It is the exact content of the Bridging Machine: a well-defined computational process, seeded at $\{1\}$, in which fertility is the exclusive property of twin prime witnesses, and in which BC is precisely the condition that guarantees the process never runs out of fertile elements.

8.5. The heuristic weight of simplicity.

Remark 8.11 (Why BM's simplicity is evidence for BC). The Bridging Machine is extraordinarily simple. Its complete description fits in three lines: seed $Q = \{1\}$; repeatedly dequeue the smallest witness v ; test every sum $u + v$ for $u \in A$. There are no parameters, no special cases, no analytic input — only a primality test to classify each sum $u + v$ as a witness or not. And yet this machine, from a single seed, generates every natural number.

This simplicity bears heuristic weight that should not be underestimated. The question BC asks is: does this machine ever stop? And the structural answer — the one BM makes viscerally apparent — is: what conceivable arithmetic conspiracy could cause it to?

For BM to halt, there must exist a threshold V beyond which no pair of witnesses $u, v \leq V$ sums to a new witness $w > V$. This is not merely a gap in the witness set; it is a global failure of additive closure at every pair simultaneously. The witnesses below V must collectively fail to bridge V in every combination.

The defect form analysis (Section 4) tells us that witnesses are structurally sparse — and yet sparsity alone is not enough to cause this failure. The three Balestrieri families block most integers, but they do so with correlated arithmetic structure that has, for every threshold examined, always left enough compatible pairs to bridge it. The probability that all pairs fail simultaneously at any given threshold is not merely small — it is, by direct probabilistic estimate, so vanishingly small as to be beyond any reasonable notion of coincidence. The authors invite anyone who believes otherwise to provide a concrete estimate.

It would be extraordinarily surprising if a machine of such transparent simplicity, which has covered every natural number examined without exception, were to fail at some remote threshold — not because the witnesses thin out (they always thin out), but because their

additive structure, which has been self-reinforcing at every scale, suddenly ceases to be. The structural tension of Section 4 resolves at every scale we can see. There is no known mechanism by which it would fail to resolve at scales we cannot.

This is not a proof. But it is the kind of evidence that mathematics takes seriously: a machine of irreducible simplicity, apparently running forever, with no visible reason to stop.

9. THE GAP MACHINE

The Bridging Machine builds $\mathcal{W}$ outward from ω , running forever if and only if BC is true. The Gap Machine is its structural dual: it verifies the gap bound on consecutive witnesses. Under WDC it halts (every pair is decomposable); GC false forces it to loop forever on the first counterexample.

9.1. Definition of the Gap Machine.

Definition 9.1 (Gap Machine, GM). The *Gap Machine* processes consecutive witness pairs in order. Its state is $(n, w_n, w_{n+1}, \text{status})$ where $w_n < w_{n+1}$ are consecutive witnesses and status is one of SEARCHING, VERIFIED, or HALT.

Initialise with $n = 1$, $w_1 = \omega = 1$, w_2 the next witness.

Each *step* at state $(n, w_n, w_{n+1}, \text{SEARCHING})$:

- (1) Search for a decomposition $w_{n+1} = u + v$ with $u, v \in \mathcal{W}$ and $u \leq v \leq w_n$.
- (2) If found: set status to VERIFIED; advance to $(n+1, w_{n+1}, w_{n+2}, \text{SEARCHING})$.
- (3) If not found: loop — return to SEARCHING on the same (n, w_n, w_{n+1}) indefinitely.

The machine *halts* if it successfully verifies every consecutive pair and $\mathcal{W}$ is exhausted (TPC is false). If $\mathcal{W}$ is infinite and every pair is verified, the machine runs forever in the VERIFIED/ADVANCE cycle. If any pair fails, the machine loops forever on that pair.

Remark 9.2 (Why running forever is the failure case). Running forever on step 3 is not a timeout or an error — it is the precise computational signature of a GC counterexample. If $w_{n+1} > 2w_n$, then any $u + v = w_{n+1}$ with $u \leq v$ requires $v > w_n$, so no valid decomposition exists within $\mathcal{W} \cap [1, w_n]$. The machine searches indefinitely and finds nothing. The non-termination is permanent because the decomposition is impossible, not because the search is incomplete.

Conversely, if $w_{n+1} \leq 2w_n$ (GC holds at this pair), a valid decomposition is found and the machine halts on that pair, then advances. GC true means every pair is resolved and GM halts overall.

9.2. The halting theorem.

Theorem 9.3 (GM halting theorem). (1) (Under WDC.) *For every consecutive pair (w_n, w_{n+1}) , WDC supplies a decomposition $u + v = w_{n+1}$ with $u, v \in \mathcal{W}$ and $u \leq v$. Consecutiveness forces $v \leq w_n$, so GM verifies every pair and halts. (GC follows from WDC via Theorem 5.3 and is not needed as a separate hypothesis.)*

(2) (Unconditional.) *If GC is false, there exists a consecutive pair with $w_{n+1} > 2w_n$. No decomposition $u + v = w_{n+1}$ can have $v \leq w_n$, so GM finds nothing and runs forever on this pair.*

In particular: WDC implies GM halts; GC false implies GM runs forever.

Proof. (1) Under WDC, w_{n+1} decomposes as $u + v = w_{n+1}$ with $u \leq v$ and $u, v \in \mathcal{W}$. Since w_{n+1} is the *next* witness after w_n , every witness strictly below w_{n+1} satisfies $v \leq w_n$. So the decomposition automatically has $v \leq w_n$, and GM finds it and advances.

(2) If $w_{n+1} > 2w_n$: any $u + v = w_{n+1}$ with $u \leq v$ requires $v \geq w_{n+1}/2 > w_n$. But the only witness $\geq w_{n+1}$ is w_{n+1} itself, forcing $u = 0$ — not a witness. No valid decomposition $u + v = w_{n+1}$ with $u, v \in \mathcal{W}$ and $v \leq w_n$ exists. GM loops. $\square$

9.3. Duality with the Bridging Machine.

Theorem 9.4 (BM–GM duality). *The Bridging Machine and the Gap Machine are structurally dual:*

	BM	GM
<i>Seed</i>	$\{1\}$	(ω, w_2)
<i>Direction</i>	<i>builds outward</i>	<i>verifies inward</i>
<i>Conjecture</i>	<i>BC</i>	<i>WDC (GC false $\Rightarrow$ loop)</i>
<i>Certificate</i>	<i>new witness found</i>	<i>decomposition found</i>
<i>Conjecture true</i>	<i>runs forever</i>	<i>halts</i>
<i>Conjecture false</i>	<i>halts</i>	<i>runs forever</i>

Remark 9.5 (Asymmetry in what is proved). The duality in the table is the intended structural picture. However, neither direction is a full biconditional. For GM, Theorem 9.3 gives $\text{WDC} \Rightarrow \text{GM halts}$ and $\text{GC false} \Rightarrow \text{GM loops}$; the converse ($\text{GM halts} \Rightarrow \text{WDC}$) is not claimed. For BM, only one direction is proved: $\text{BC true} \Rightarrow \text{BM runs forever}$

(Theorem 8.4). The full BM equivalence is stated as Claim 8.6; the converse is deferred to future work.

The Lean 4 formalisation (Aristotle, run 1) confirms this picture: `gc_implies_gm_advances` requires WDC as a hypothesis, not GC alone. GC bounds the gap but does not produce the decomposition; WDC does both.

Remark 9.6 (Two machines, one structure). BM and GM are two views of the same underlying arithmetic. BM asks: *can witnesses always be extended?* GM asks: *can witnesses always be decomposed?* Under WDC (which follows from BC via Theorem 6.6), these questions are equivalent: extension and decomposition are inverse operations on $\mathcal{W}$.

The duality is precise. BM’s halting condition (queue empties) is the exact negation of BC. GM’s infinite loop (no decomposition found) is the exact negation of GC. Since GC follows from BC (Theorem 5.4), BC true implies BM runs forever *and* GM never loops — both machines confirm the structure of $\mathcal{W}$ simultaneously, from opposite directions.

The two machines together give a complete computational picture of the Structural Conjecture: PNT forces the arithmetic that keeps BM running and GM advancing, with neither machine ever stopping for the wrong reason.

Part 3. Line 2: Channel Exhaustion as Analytic Bridge

10. LINE 2: CHANNEL EXHAUSTION AS ANALYTIC BRIDGE

The additive structural approach of Section 6 is self-contained: BC $\Rightarrow$ TPC requires no analytic input. This section presents a second, independent line of evidence that is *not* a parallel proof route but a diagnostic tool — it makes precise where the structural argument hands off to analysis, and identifies the question that analytic methods must answer.

Conjecture 10.1 (Channel Exhaustion). *If $\mathcal{W}$ is finite with largest element $w_{\max}$, then for every $m > w_{\max}$, both $L_m = 6m - 1$ and $H_m = 6m + 1$ are composite.*

Remark 10.2. The conjecture asserts that a finite witness set forces structural exhaustion of witness-compatible channel configurations: no channel index above $w_{\max}$ supports even a single prime in both channels simultaneously.

The natural approach via a finite covering — showing that some set of small primes blocks all residue classes — is unavailable. No finite set of primes can cover all residue classes, since the uncovered fraction

$\prod_p(1 - 2/p)$ is always positive. A direct proof of Conjecture 10.1 must account for the $\{0, 2, 3\} \pmod{5}$ residue classes, where $m \notin \mathcal{W}$ implies only that the two channel values are not *simultaneously* prime — one may still be an individual prime.

Channel exhaustion sits nearer to classical analytic questions than WDC or BC, but lacks the direct additive closure structure that drives the main argument. Closing it likely requires analytic input — the Prime Number Theorem at minimum.

Theorem 10.3 (Channel exhaustion $\Rightarrow \mathcal{W}$ finite $\Rightarrow \mathbb{P}$ finite). *Conjecture 10.1 implies: if $\mathcal{W}$ is finite then $\mathbb{P}$ is finite.*

Proof. Assume $\mathcal{W}$ is finite and Conjecture 10.1 holds. Let $N = 6w_{\max} + 1$. Let $p > N$ be any prime; then $p > 3$, so by Lemma 2.3 there exists $m \geq 1$ with $p = L_m$ or $p = H_m$. Since $p > 6w_{\max} + 1$, we have $m > w_{\max}$. By channel exhaustion, both L_m and H_m are composite — contradicting p being prime. Hence no prime exceeds N , so $\mathbb{P}$ is finite. $\square$

Remark 10.4 (Why the implication is non-trivial). It might seem that $\mathcal{W}$ finite trivially implies $\mathbb{P}$ finite: if there are only finitely many twin prime witnesses, there are only finitely many twin prime pairs, and twin primes are primes. This argument is wrong.

Finitely many twin primes does *not* imply finitely many primes. A prime p such that neither $p - 2$ nor $p + 2$ is prime — an isolated prime — is a valid prime that contributes to $\mathbb{P}$ without contributing to $\mathcal{W}$. Isolated primes are conjectured to be infinite in number regardless of TPC.

The non-trivial content of Theorem 10.3 is that $\mathcal{W}$ finite, combined with channel exhaustion, forces *all* primes — twin or isolated — to be bounded. The mechanism is Lemma 2.3: every prime $p > 3$ occupies a channel index. If all channel indices above $w_{\max}$ have both values composite, no prime of any kind — twin or not — can exist there.

Remark 10.5 (Euclid's veto). The full implication chain is:

$$\mathcal{W} \text{ finite} \xrightarrow{\text{Conj. 10.1}} \mathbb{P} \text{ finite} \xrightarrow{\text{Euclid}} \perp.$$

Therefore $\mathcal{W}$ is infinite — TPC — unconditionally once Conjecture 10.1 is proved. The consequent ($\mathbb{P}$ finite) is necessarily false by Euclid; the antecedent ($\mathcal{W}$ finite) is necessarily false under TPC. The logical structure is: $(A \Rightarrow \perp) \equiv \neg A$. The proof of TPC via this route is not a proof *about* $\mathbb{P}$ being finite; it is a demonstration that the premise $\mathcal{W}$ finite cannot hold in any consistent instance.

Part 4. The Structural Conjecture

11. THE STRUCTURAL CONJECTURE

The preceding sections develop the algebraic infrastructure of the witness set and two independent lines of evidence toward TPC. This section states the central open conjecture that the paper is built to motivate.

Conjecture 11.1 (The Structural Conjecture).

$$\text{PNT} \implies (\text{WDC} \wedge \text{BC}).$$

Remark 11.2 (What the conjecture claims). The Prime Number Theorem asserts that $\pi(x) \sim x/\log x$ — a purely analytic, asymptotic statement about the density of primes. The Structural Conjecture asserts that this single analytic fact is strong enough to force both additive closure properties of $\mathcal{W}$: that every witness decomposes (WDC) and that witnesses bridge every threshold (BC).

This is *not* a density claim about $\mathcal{W}$ itself. It is a structural sufficiency claim: PNT provides sufficient analytic raw material — guaranteeing asymptotic prime supply across both channels — to force the witness set into additive closure. PNT is the minimal large-scale analytic statement guaranteeing asymptotic prime supply across both channels, making it the natural candidate for forcing witness closure if such closure is analytically determined at all.

Remark 11.3 (Why this may bypass the parity obstruction). Classical analytic approaches to TPC encounter the parity obstruction: sieve methods cannot distinguish between numbers with an even and an odd number of prime factors, and twin primes require primality in both channels simultaneously. The structural approach inverts the question. Rather than asking “do $6m - 1$ and $6m + 1$ avoid all prime factors?” it asks “does $\mathcal{W}$ have sufficient additive closure to be infinite?” The framework avoids direct engagement with parity obstructions because it does not attempt to detect primality via sieve cancellation. If $\text{PNT} \implies \text{WDC}$, the parity problem is bypassed entirely: the question of primality is never asked directly.

Remark 11.4 (The structural tension as engine of the Structural Conjecture). Observation 4.12 identifies the heuristic mechanism behind the left arrow of the central thesis.

The defect forms impose two competing pressures on any witness sum $w = u + v$. BC requires v to be close enough to the threshold that the sum just clears it. The defect structure requires u and v to be algebraically compatible as witnesses — a constraint that generically

favours separation rather than proximity to $w/2$. Twin primes arise from the resolution of this tension.

The role of PNT is to guarantee that the tension is always resolvable: that for any threshold V , the supply of witnesses below V is sufficient — in density and arithmetic distribution — that some compatible pair (u, v) exists whose sum is a new witness above V . PNT controls the density of witnesses (via the density of primes in both channels), and it is this density that determines whether the algebraic compatibility constraints can always be met.

If PNT provides enough witness density, the two pressures are always simultaneously satisfiable, and BC holds. Under WDC, $BC \Leftrightarrow TPC$ (Theorem 6.6), so the Structural Conjecture would follow. The left arrow is then not an independent analytic conjecture but a consequence of PNT acting through the defect structure — the algebraic engine that makes the tension productive rather than obstructive.

Remark 11.5 (Relationship to channel exhaustion). The two lines of evidence are not independent routes to the same destination. PNT may be precisely what is needed to close Line 2 (channel exhaustion), and channel exhaustion may be what forces Line 1 (additive closure). The Structural Conjecture proposes that the bridge between the analytic world and the additive structural world exists and is crossed by PNT alone.

11.1. The full implication chain. The central thesis of this paper, fully assembled:

$$\text{PNT} \xRightarrow{\text{Conj. 11.1}} (\text{WDC} \wedge \text{BC}) \xRightarrow{\text{Thm 6.5}} \text{TPC} \xRightarrow{\text{Euclid}} \mathbb{P} \text{ infinite.}$$

The right two arrows are proved. The left arrow is the Structural Conjecture.

The channel exhaustion route provides a parallel conditional path:

$$\text{PNT} \Rightarrow \text{Conj. 10.1} \xRightarrow{\text{Thm 10.3}} (\mathcal{W} \text{ finite} \Rightarrow \mathbb{P} \text{ finite}) \xRightarrow{\text{Euclid}} \mathcal{W} \text{ infinite.}$$

11.2. Immediate corollaries of the Structural Conjecture.

Corollary 11.6 (TPC). *The Structural Conjecture implies the Twin Prime Conjecture.*

Proof. $\text{PNT} \Rightarrow \text{BC}$ by the Structural Conjecture; $\text{BC} \Rightarrow \text{TPC}$ by Theorem 6.5. $\square$

Corollary 11.7 (Equivalence of structural conjectures). *Under the Structural Conjecture, $\text{WDC} \Leftrightarrow \text{BC} \Leftrightarrow \text{TPC}$.*

Proof. $BC \Rightarrow TPC$ is Theorem 6.5. $TPC \Rightarrow BC$ follows from Theorem 6.6 once WDC is established by the Structural Conjecture. $WDC \Rightarrow BC$ follows from $WDC + TPC \Rightarrow BC$ (Theorem 6.6) once TPC is known. $\square$

Corollary 11.8 (Bertrand’s postulate for twin prime pairs). *The Structural Conjecture implies: for every twin prime pair $(p, p + 2)$, the next twin prime pair $(q, q + 2)$ satisfies $q \leq 2p + 2$.*

Proof. Let w_n be the witness for $(p, p + 2)$, so $p = 6w_n - 1$. The Structural Conjecture gives WDC, and $WDC \Rightarrow GC$ (Theorem 5.3), so $w_{n+1} \leq 2w_n$. Hence $q = 6w_{n+1} - 1 \leq 12w_n - 1 = 2(6w_n - 1) + 1 = 2p + 1 < 2p + 2$. $\square$

Remark 11.9. Corollary 11.8 is the witness analogue of Bertrand’s postulate ($p_{n+1} < 2p_n$ for primes), applied to twin prime pairs rather than individual primes. Classical sieve methods have not produced an explicit finite gap bound between consecutive twin prime pairs; the structural approach yields one as a direct corollary of WDC alone. The bound $q \leq 2p + 1$ is tight in the sense that GC is verified to 10^5 witnesses with equality approached but not achieved.

11.3. Open questions.

- (1) Can WDC be established from PNT alone, with BC following from Theorem 6.6?
- (2) Is PNT the minimal analytic hypothesis, or does the Structural Conjecture hold under a weaker assumption?
- (3) Does proving $GC \Rightarrow WDC$ (Conjecture 6.8) reduce the Structural Conjecture to $PNT \Rightarrow GC$ — a simpler metric statement?
- (4) Can channel exhaustion (Conjecture 10.1) be derived from PNT, providing an analytic proof of Line 2?

12. CONCLUSION

This paper develops a structural reformulation of the Twin Prime Conjecture in which the remaining obstruction is isolated into a precise additive closure conjecture. The channel representation (Section 2) places every prime above 3 in one of two arithmetic channels. The Multiplexing Identity (Section 3) provides the algebraic engine connecting witness addition to channel arithmetic. Algebraic constraints (Section 4) establish hard, unconditional bounds on witness configurations. The Gap Conjecture (Section 5) is derived as a metric corollary of the structural conjectures, not assumed as a primitive.

Two independent lines of evidence point toward TPC. Line 1 (Section 6) is purely algebraic: the Bridging Conjecture implies TPC unconditionally, and under WDC, $BC \Leftrightarrow TPC$. Line 2 (Section 10) connects the additive structure of witnesses to the broader distribution of primes via channel exhaustion, isolating precisely where analytic tools are needed.

The Structural Conjecture (Section 11) proposes that the Prime Number Theorem alone forces both WDC and BC. If true, this would yield TPC without direct engagement with the parity obstruction that has blocked classical sieve-theoretic approaches.

What remains: a proof that PNT implies additive closure of $\mathcal{W}$ — or a proof of channel exhaustion from PNT, which would close Line 2 and imply TPC via Euclid. The framework reduces the infinitude of twin primes to a precise structural closure principle on the witness semigroup.

The channel exhaustion conjecture is the interface where the structural programme hands off to analysis. Proving it is the invitation extended to the analytic community.

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APPENDIX: LOGICAL DEPENDENCY MAP

The following table records all implications proved or conjectured in this paper, together with their status. Theorem numbers are given once numbering stabilises.

Implication	Status	Condition	Reference
<i>Proved, unconditional</i>			
$BC \Rightarrow TPC$	Proved	—	Thm 6.5
$BC \Rightarrow GC$	Proved	—	Thm 5.4
$WDC \Rightarrow GC$	Proved	—	Thm 5.3
Channel Exhaustion $\Rightarrow (\mathcal{W} \text{ fin.} \Rightarrow \mathbb{P} \text{ fin.})$	Proved	—	Thm 10.3
$\mathbb{P} \text{ finite} \Rightarrow \perp$	Proved	—	Euclid
<i>Proved, conditional</i>			
$WDC + TPC \Rightarrow BC$	Proved	WDC, TPC	Thm 6.6
<i>Conjectured</i>			
$PNT \Rightarrow (WDC \wedge BC)$	Open	—	Conj 11.1
$GC \Rightarrow WDC$	Open	—	Conj 6.8
Channel Exhaustion	Open	—	Conj 10.1
WDC	Open	—	Conj 6.1
BC	Open	—	Conj 6.3
<i>Corollary (follows from either structural conjecture)</i>			
GC	Open	WDC or BC	Conj 5.1, Thms 5.3, 5.4

The main chain (right arrow proved; left arrow conjectured):

$$PNT \dashrightarrow WDC \wedge BC \longrightarrow TPC \longrightarrow \mathbb{P} \text{ infinite.}$$

The channel exhaustion route:

$$PNT \dashrightarrow \text{Ch. Exhaustion} \longrightarrow (\mathcal{W} \text{ fin.} \Rightarrow \mathbb{P} \text{ fin.}) \longrightarrow \mathcal{W} \text{ infinite.}$$

Metric shadow:

$$WDC \longrightarrow GC \longleftarrow BC, \quad GC \dashrightarrow WDC \text{ (open).}$$

Arrow conventions: solid $\longrightarrow$ = proved unconditionally; dashed $\dashrightarrow$ = conjectured; double $\Longleftrightarrow$ = proved equivalence under stated condition.

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